PORTFOLIO CASE STUDY · POWER BI

Agricultural production and food security monitor

Staple-crop output across 25 countries grew 65% since 1990, but yield stopped improving in 2014. I measured the yield gap and where closing it would add the most food.



Taiwo Ajayi
Data analyst · Power BI developer · StarTechOne Nigeria Limited



DOMAIN
Agriculture and food security

DATA
25 countries, 11 crops, 17 survey years (1990-2022)

MODEL
Star schema: 1 fact, 4 dimensions, measures table

REPORT
17 pages: 13 story, 3 bento dashboards, 1 drill-through

Power BI · DAX · Data modelling · Benchmarking · Public data · Storytelling

THE PROJECT

The challenge

Policy and investment decision-makers needed to know whether staple-crop production is still growing, whether growth comes from yield or more land, and where raising yields to the regional best would add the most food.

The data

An FAO-style production extract: production, area harvested and yield by country, crop and survey year (every second year). 14.6% of records are estimates and 252 values are blank. Totals add up all 17 survey years, so trend pages show one year at a time.

My approach

1. Profiled the workbook and checked every headline number against the source.
2. Fixed the regional benchmark so it is a true 75th percentile of country yields.
3. Removed a wrong hg/ha conversion that shrank the potential-gain figure 10,000 times.
4. Added 12 measures (shares, survey-year values, gap counts, potential gain %).
5. Built 13 story pages and 3 bento dashboards with synced Year, Region and Crop filters.

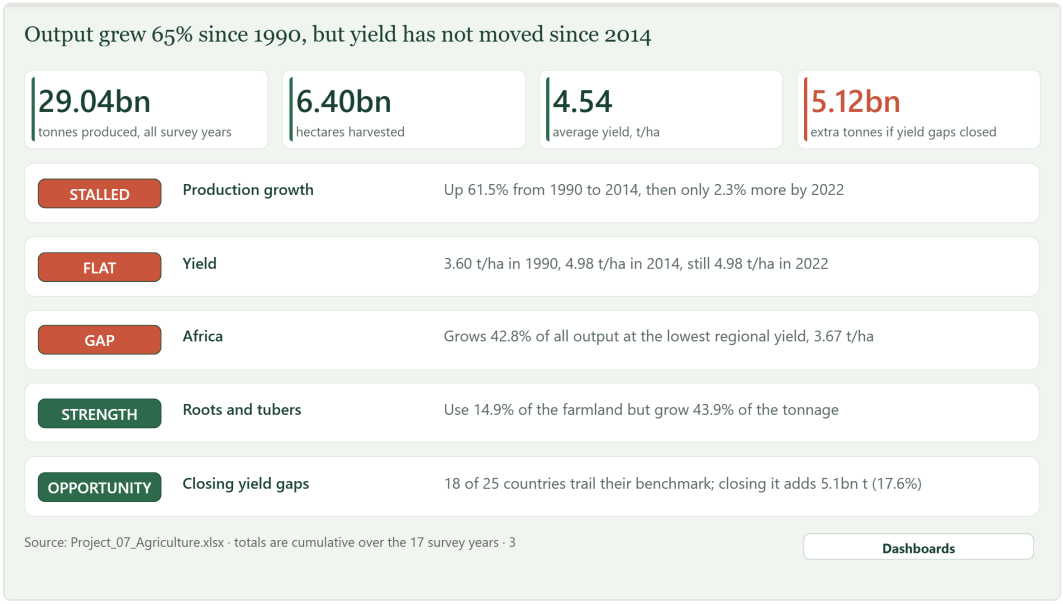
Headline numbers

29.0bn t
production, all survey years

4.54 t/ha
average yield

18 of 25
countries below benchmark

+5.1bn t
if yield gaps closed (+17.6%)



The scorecard page: headline KPIs and verdicts on each finding.

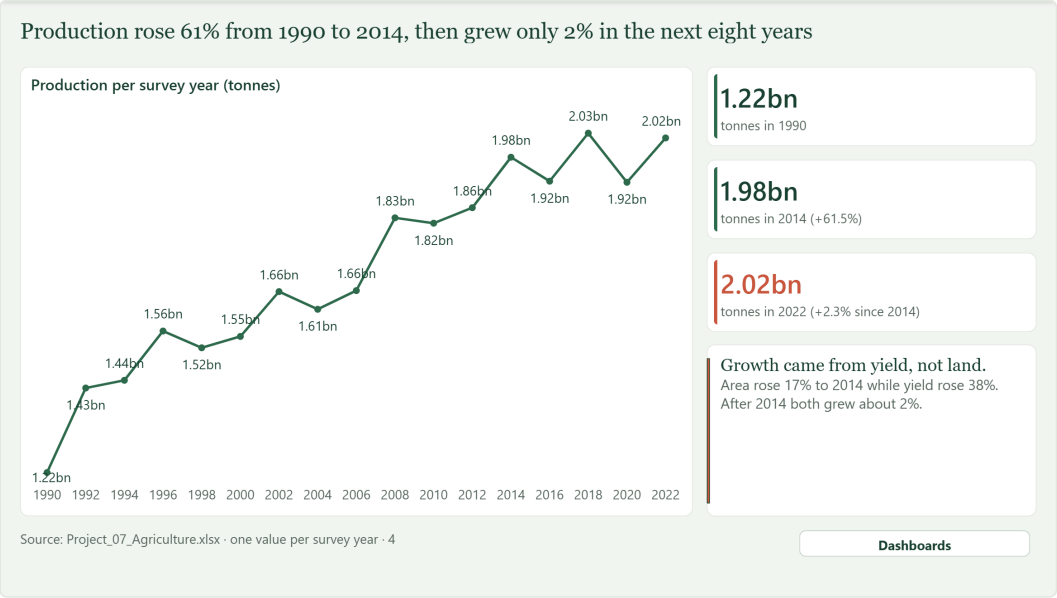
RESULTS

Key insights

Growth has stalled	Production rose 61.5% from 1990 to 2014, then only 2.3% by 2022.
Yield is flat since 2014	3.60 t/ha in 1990, 4.98 in 2014, still 4.98 in 2022.
Africa grows most, yields least	42.8% of all output at the lowest regional yield, 3.67 t/ha.
Roots and tubers are efficient	14.9% of the land, 43.9% of the tonnage.
Six countries hold half the upside	Philippines, Argentina, Brazil, Malawi, Uganda and India.

Recommendations

1. Close Africa's yield gap: seeds, fertiliser and extension.
2. Restart yield growth with research and irrigation in the six largest-gain countries.
3. Back roots and tubers with storage and processing.



A story page: the title states the finding and the chart shows the evidence.

Three interactive dashboards

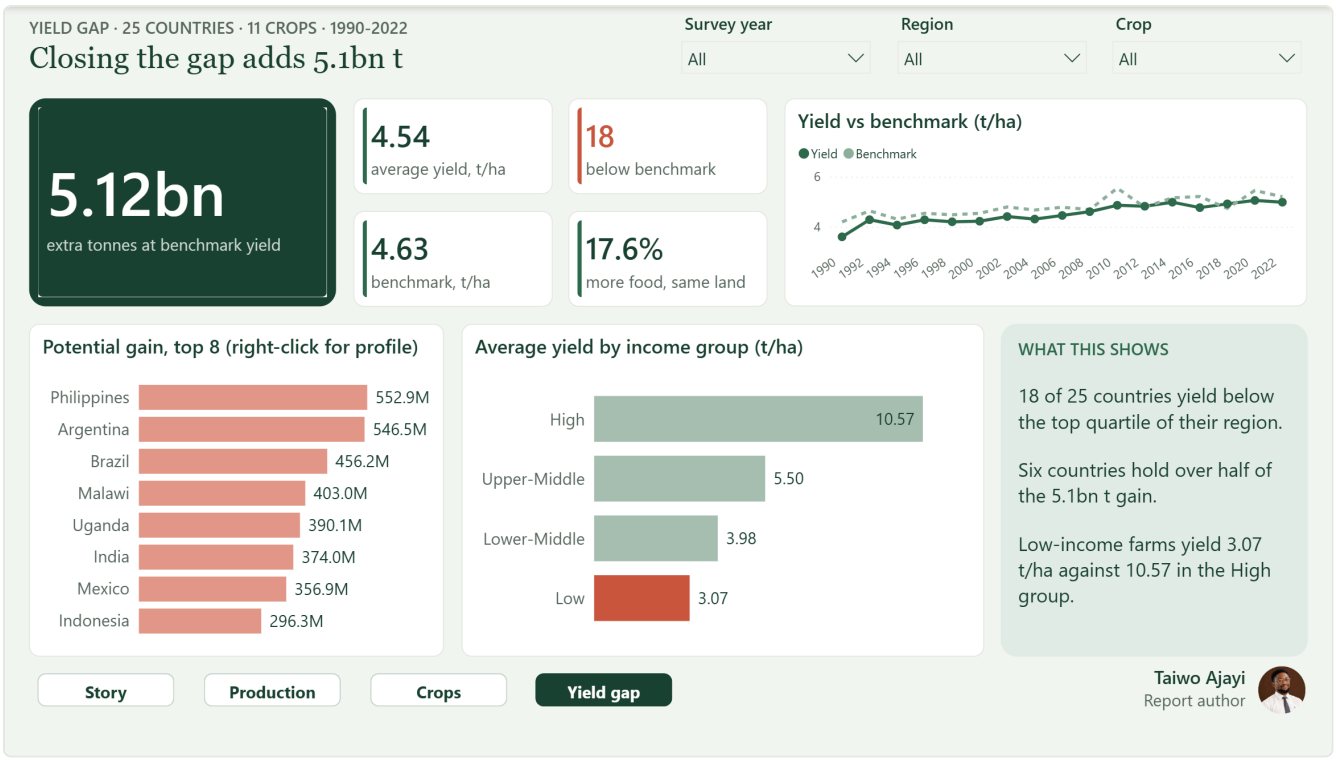
Each dashboard has synced filters, a tab bar to move between them, and a drill-through to a detail page.



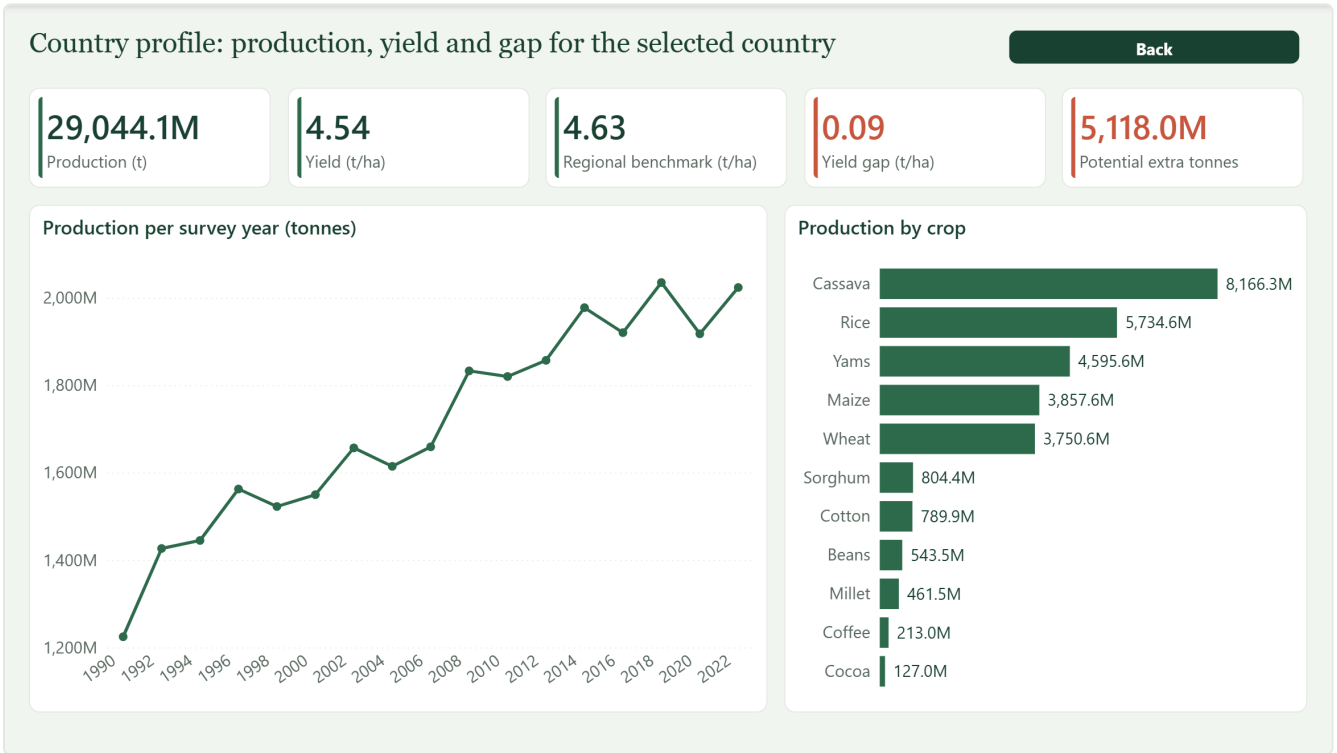
D1 Production overview. Output, trend, regions and top producers



D2 Crop analysis. Crops, crop families and yields



D3 Yield and gap. Yield vs benchmark and potential gain



Country profile. Hidden drill-through page: right-click a bar on a dashboard to open it

QUALITY AND SKILLS

What I fixed, and what this shows

Errors I found and corrected in the original analysis

- Corrected swapped region shares on the original insights page.
- Replaced a mislabelled "top-quartile" benchmark with a true 75th percentile.
- Removed a divide-by-10,000 error in potential production gain.

Skills demonstrated

- Data profiling and validation before reporting
- Star-schema modelling and DAX measure design
- Correcting flawed measures and documenting why
- Insight-led storytelling: one finding per page
- Executive dashboard design with drill-through
- Clear documentation for readers and maintainers

Tools

- Power BI Desktop (PBIP project format)
- DAX and Power Query (M)
- TMDL and PBIR source files
- Excel source data

Deliverables

- 17-page Power BI report
- Full report PDF (every page)
- Detailed report guide



The data notes page: every correction and data limit is stated in the report itself.